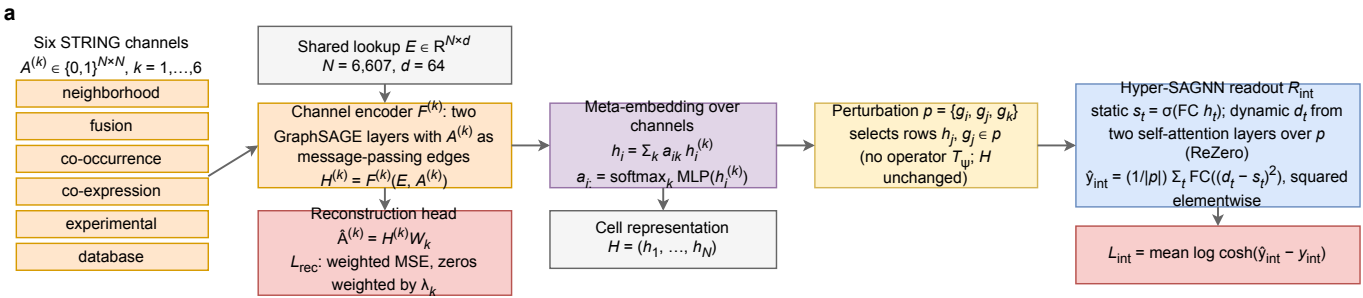
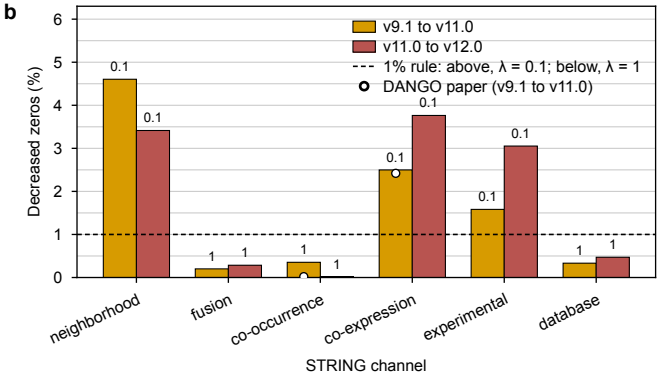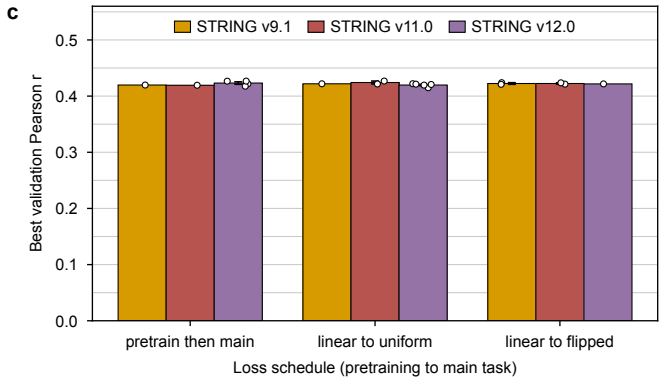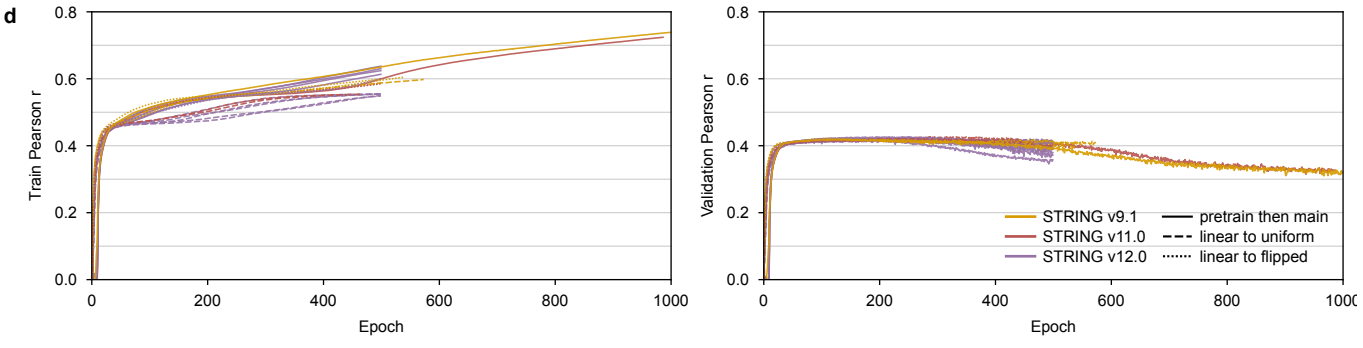
Objective at epoch e: $L = \alpha_e L_{rec} + (1 - \alpha_e) L_{int}$; $\lambda_k = 0.1$ if the channel's decreased zeros exceed 1%, else 1 (panel b). Schedules with transition epoch 10: pretrain then main, $\alpha_e = 1$ for $e <$ 10, then 0; linear to uniform, α_e 1 → 0.5 by epoch 10, then 0.5; linear to flipped, α_e 1 → 0 by epoch 10, then 0.